
Master Research Proposal

This document is to be filled by the student in cooperation with the supervisor /s.

General Information

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Research Proposal Title: Vision-Based Extraction and LLM-Powered Summarization of TV News Tickers

Main Supervisor: Prof. Dr. Muhammed Shahzad

Abstract

News videos contain continuous ticker text that delivers short, real-time information updates. Automatically extracting and summarizing this ticker content is challenging due to low resolution, motion artifacts, color variations, and rapid text transitions. At the same time, recent advances in Large Language Models (LLMs) offer strong capabilities for summarizing short but dense textual information.

This Master's thesis proposes an end-to-end system that detects and isolates the news ticker region in broadcast videos, extracts its textual content using Optical Character Recognition (OCR), and applies multiple LLMs to generate summaries. A custom dataset will be created to evaluate the quality of extracted text and the performance of different LLMs. The study will benchmark models on accuracy, faithfulness, latency, computational cost, and robustness to OCR noise.

The goal is to provide a systematic analysis of how modern LLMs perform when summarizing OCR-extracted real-world ticker text, and to identify which models are most suitable for this domain.

Introduction

Broadcast news videos typically include a dynamic ticker displayed at the bottom of the screen. This ticker contains concise yet highly informative textual updates. Automatically processing this text is difficult because the ticker appears in low resolution, moves horizontally, and often suffers from compression artifacts and complex backgrounds.

Recent progress in video processing and OCR has enabled more reliable extraction of text from video content. In parallel, Large Language Models (LLMs) have shown strong performance in

abstractive summarization tasks. Despite these advancements, there is limited research combining these components into a unified pipeline specifically focused on news tickers.

This thesis aims to fill this gap by developing an integrated system capable of detecting ticker regions in videos, extracting text using OCR, and generating summaries using multiple LLMs. The work will include building a ground-truth dataset and evaluating different models using standard summarization metrics and human assessment. The outcome will be a systematic study of LLM summarization performance under real-world OCR conditions.

Research Objectives and Approach

Objectives

1. Develop a robust method to detect and segment the news ticker region in video frames.
2. Extract ticker text using OCR and refine the text through post-processing techniques.
3. Design and construct a dataset of ticker text paired with ground-truth summaries.
4. Apply several LLMs to generate summaries of the extracted text.
5. Benchmark the performance of the LLMs using automated and human evaluation metrics.
6. Analyze tradeoffs in accuracy, robustness to OCR noise, latency, and computational cost.

Approach

This thesis will be executed through the following steps:

1. **Ticker Detection:**
Use computer vision techniques (color segmentation, edge detection, or trained detection models) to identify the ticker region in each video frame.
2. **Text Extraction:**
Apply OCR tools such as Tesseract or EasyOCR to convert the detected region into machine-readable text. Perform cleaning, normalization, and duplicate removal.
3. **Dataset Creation:**
Collect a set of news videos, manually transcribe ticker text for ground truth, and create reference summaries.
4. **Summarization:**
Use multiple LLMs (e.g., open-source and API-based models) to produce summaries under consistent prompting conditions.
5. **Evaluation:**
Compare system outputs using ROUGE, BERTScore, faithfulness checks, human evaluation, execution time, and cost metrics.
6. **Analysis:**
Identify which LLM performs best for this specific real-world task and discuss limitations and future improvements.

Signatures

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